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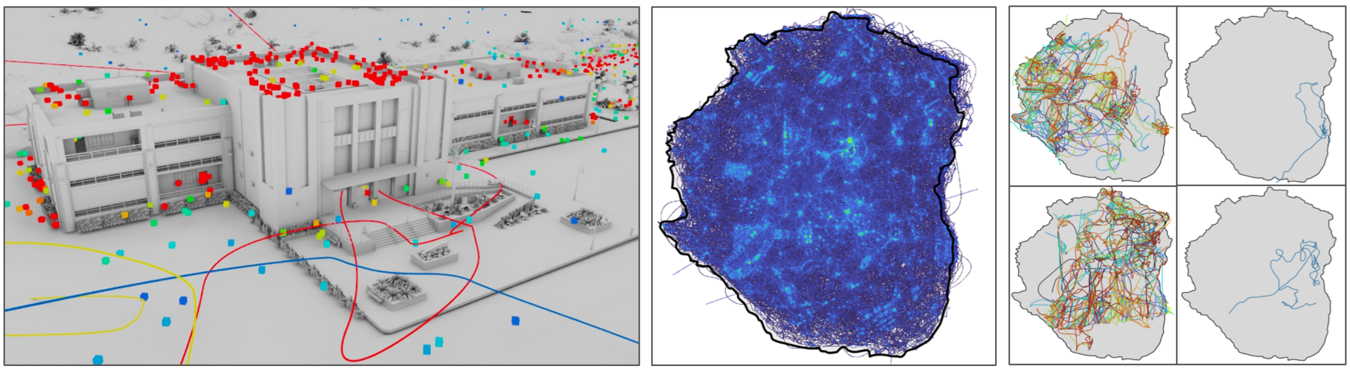

Figure 1: Left: The Arsenal area on the Caldera map in Call of Duty®: Warzone™, overlaid with player movement paths from a single match and their endpoint locations. (©Activision Publishing, Inc.) Middle: A heatmap visualization of aggregated player trajectories across multiple matches, highlighting high-traffic zones. Right: Movement dynamics of the player paths for different matches and players.

Abstract

Understanding player behavior, movement, and dynamics is crucial for uncovering how players interact with game environments, optimize strategies, and engage in competitive scenarios, offering insights that inform game design, AI development, and human interaction. We present a spatial-temporal analysis of player movement, path trajectories, and behavioral strategies on the Call of Duty: Warzone Caldera map using an open-source telemetry dataset. We characterize player motion through kinematic metrics such as distance, entropy, acceleration, and jerk to capture fine-grained movement dynamics under changing game conditions. Our analysis highlights the interplay and behavioral patterns across skill levels and player roles by comparing the highest and lowest-ranked players, revealing strategies linked to path optimization and survival.

CCS Concepts

• **Human-centered computing** → *Visualization*; • **Information systems** → **Massively multiplayer online games**; • **Applied computing** → **Computer games**; • **Software and its engineering** → **Interactive games**; • **Computing methodologies** → *Motion processing*.

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Keywords

Player Behavior Modeling, Player Dynamics, Movement, Player Patterns, Kinematic Analysis, Battle Royale

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1 Introduction

Call of Duty®: Warzone™ is one of the most widely played battle royale games, attracting millions of players worldwide with its fast-paced gameplay, dynamic environments, and competitive structure. This is a free-to-play first-person shooter that challenges players to outlast opponents in competitive matches that combine tactical movement, resource management, and real-time combat. Warzone™ maps are large-scale designed environments that blend urban, rural, and natural terrain to support diverse strategic gameplay, requiring players to continuously adapt their movement, positioning, and tactical decisions. Player behavior is shaped by environmental features of the map, including topographically varied terrain, the spatial distribution of loot, the density of player encounters, and the shifting boundaries of the playable zone. These elements collectively influence player decision making, movement trajectories, and engagement strategies throughout the match. The complexity of player interactions, coupled with the competitive nature of the gamemode, generates diverse play patterns that vary by skill level and in-game contexts.

Game Data Science [El-Nasr et al. 2021] is an emerging interdisciplinary field that combines game design, behavioral analytics, machine learning, and data visualization to extract insights from player interactions, optimize game experiences, and inform both development and research in interactive systems. Large-scale analysis of player behavior is essential for understanding how players adapt, optimize strategies, and compete within each match. This provides valuable insights to improve game design and advance research in artificial intelligence, human behavior modeling, and digital interaction.

The Caldera dataset [Activision 2024], released by Activision, provides an open-source, comprehensive collection of player telemetry from Call of Duty®: Warzone's™ Caldera map, comprising granular data on player movement, player life status, player endpoints, and map geometry. The data enables spatial-temporal modeling of player trajectories, supporting investigations into player behaviors. The dataset includes a full 3D representation of the Caldera map provided in a Universal Scene Description (USD) format, capturing terrain, structures, and environmental features with high spatial fidelity (Figure 1).

This paper presents a framework for analyzing player behavior in large-scale multiplayer game environments using spatial-temporal telemetry data. We introduce a method for reconstructing continuous player trajectories from sparse and noisy breadcrumb data, enabling the identification of key gameplay events. Our approach incorporates a set of interpretable metrics for quantifying player movement patterns and behaviors. Additionally, we provide visualizations and case studies that demonstrate how these metrics can be used to understand differences in player behavior across skill levels and match outcomes. These findings help to better understand player behavior in large-scale competitive battle royale-style games, which can be used to aid advancements in skill-based matchmaking and assist in replicating realistic human movement behavior in AI agents.

2 Background

Call of Duty®: Warzone™ (CoD) is a free-to-play battle royale style game developed by Infinity Ward and published by Activision alongside their title Call of Duty®: Modern Warfare™. In this gamemode, players fight in a larger-scale match until all players but one are eliminated. Players have the option to deploy into solo matches or team up with others in squads for a maximum of four players per team. Solo matches can have a maximum of 120 players at a time.

CoD: Warzone™ is regularly updated with new content and several large-scale maps, each designed with distinct geographic and strategic opportunities. The map Caldera was the second large-scale unique map released to players in 2021 (Figure 2), one year after the game's initial release. Comparing this map with its predecessor, Verdansk, Caldera's vertical geography gives players more natural cover in the form of valleys, cliffs, and mountains. Caldera is separated into 15 distinct regions of interest. This includes a capital city in the south, a caldera land formation in the center, a naval base towards the north, and many more. With the number of unique locations, players face a wide variety of scenarios that require adaptive strategies to succeed.

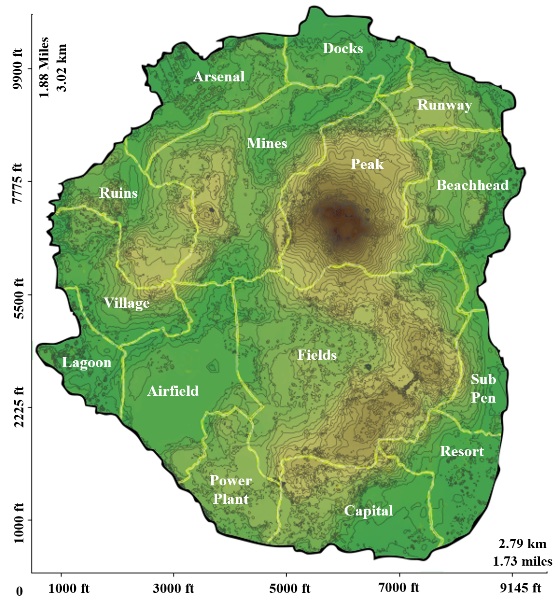

Figure 2: Topographic map of the Caldera Map in CoD Warzone featuring the 15 regions of interest.

2.1 Game Mechanics

The game starts with all players being transported into the drop plane, where players can deploy anywhere on the map. The drop plane's route changes with each match, forcing players to adjust their strategies based on this trajectory. Players will need to scour dense locations to find loot to survive longer. Loot can encompass many items in the match, including guns, lethal deployable items, tactical deployable items, killstreaks, body armor, and contracts. Most of these items can be found either in supply boxes, which have randomized loot, or buy stations, where players can buy specific items. Figure 3(a) shows an example of the full player movement in a match demonstrating the plane ride, deployment, and player motion which includes searching for loot.

To keep players engaged in combat, a circle is marked on the map that indicates a safe region away from the gas, which inflicts damage over time to players who are caught within these zones. To avoid early elimination, players must locate these safe regions and travel within the designated boundaries. This area continues to shrink until either a winner is declared or the gas completely engulfs the map. With each circle, the delay between stages decreases, giving higher difficulty to late-stage players as they must continuously traverse the map to be safe. The location of these circles varies from match to match, requiring players to redesign movement strategies or accelerate looting to remain secure. The circle mechanic also helps ensure that a single meta strategy does not form and encourages the use of the entire map throughout the entirety of the map's lifetime. Figure 1 (middle) shows this phenomena by plotting the motion of all players from multiple matches.

If a player is eliminated from the match, they have the opportunity to re-enter the game by successfully competing and winning a one-on-one showdown against another eliminated player, known

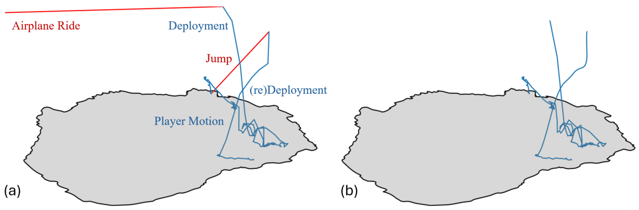

Figure 3: Player motion data from Match 0, Player 9 (final rank: 20th). (a) Original breadcrumb data showing the connected movement path. Segments in red indicate discarded data not used in the final analysis. (b) Final reconstructed player motion trajectory in blue used in the paper.

as the Gulag. The player who eliminates their opponent first can redeploy back into the match, while the defeated player must either exit the match or spectate, as they can no longer re-enter. Once the player has redeployed, their re-entry will be a spot that is determined to be safe within the current circle. The player's loot from before they were eliminated remains at the elimination location, which may encourage players to return back to that area, but they have the choice to go wherever on the map they choose. Figure 3(a) shows the airplane ride and jump we do not analyze since this motion is controlled by the game mechanics. The deployment and redeployment are controlled by the player and therefore included in analyses which can be seen in figure 3(b).

2.2 Game Telemetry and Analytics

Telemetry is a common method for unobtrusively collecting player behavior data to infer insights about player experience, improve map-making, and identify potential bugs that influence game mechanics [sei 2013; Cardamone et al. 2011; Drachen 2015]. This method of data collection enables the exact quantification of player behaviors that is not possible with observational methods. The ability to collect telemetry data is integrated into many multiplayer online games by remotely logging information about players, such as game interactions, avatar positions, and player location [Drachen 2015]. Popular games used for large-scale player telemetry studies include League of Legends, Halo, World of Warcraft, DotA 2, and Counter-Strike: Global Offensive, which have large player bases and unique game mechanics that allow additional exploration of player behavior [Afonso et al. 2019; Fernandes et al. 2018; Huang et al. 2013a; Miller and Crowcroft 2009; Xenopoulos et al. 2020a; Xenopoulos and Silva 2022]. To the author's knowledge, there has not been an academic study using telemetry data from Call of Duty®: Warzone™ to observe how the game specific mechanics impact player behaviors.

Theoretically, every action taken by a player can be logged and sent as telemetry data, but trade-offs will be required since sending massive amounts of data from every player could impact system performance and overload data storage capabilities [Drachen 2015]. Selecting metrics that are most relevant to the player behaviors of interest and the specific game type are important considerations for maximizing the informative value of the telemetry data [Canossa 2013].

2.3 Spatial and Movement Analysis in Games

Understanding video game player behavior requires analyzing data in the context of the virtual environment, the time in which the data was collected, and the player's actions preceding and following each data point. However, visualizing spatio-temporal data such as this remains a challenge for video game analysis and beyond. Traditional static visualization techniques do not allow for displaying 4D data of many players in a way that is easily decodable. Heatmaps are a common way to address this problem since they do not require highly specialized software to create and are easily understandable; however, they do require the time or density dimensionalities to be excluded [Bauckhage et al. 2014; Drachen and Schubert 2013]. Trajectory analysis improves on heatmaps, but can become cluttered, and this kind of visualization is not as accessible to non-experts [Demšar and Virrantaus 2010; Drachen and Schubert 2013]. [Wallner and Kriglstein 2020b] proposed the use of hexbins to display even more than 4 dimensions of video game data, but the additional symbology increases cognitive load and decreases readability. These challenges have led many video game visualization researchers to opt for interactive visualization interfaces [Afonso et al. 2019; Moura et al. 2011; Teng et al. 2025; Xenopoulos et al. 2022b]. Other solutions focus on clustering techniques that prioritize focus on specific insights rather than illustrating context [Ahmad et al. 2019; Wallner and Drachen 2023]. While no definitive method exists for visualizing spatio-temporal data, understanding the available techniques and their limitations enables the design and utilization of visualizations that more effectively convey the data with the appropriate context.

Player telemetry data provides metrics for analyzing in-game activity across spatial and temporal metrics. *Player movement* refers to the spatial and temporal locomotion of a player-controlled avatar within a game environment. This movement data enables the analysis of *player dynamics*, which capture the quantitative aspects of locomotion over time, including metrics such as speed, acceleration, jerk, path curvature, and directional changes. Beyond raw motion, these dynamics contribute to the identification of *player behavior*, defined as the high-level patterns and strategies inferred from movement traces—such as aggressive engagement, avoidance, camping, or exploratory traversal—often shaped by gameplay context and decision-making processes [Drachen et al. 2013; Wallner and Kriglstein 2013a].

2.4 Behavior Modeling in Digital Environments

Player behavior data is also used for modeling human behavior and creating prediction models of future player behavior [Bunian et al. 2017]. Player models that account for skill level and typical movement patterns have been used to detect uncharacteristic play activity and identify potential cheaters [Alayed 2015; Qian et al. 2022]. Creating movement models can also be used to address rendering challenges such as determining where to direct resources for best performance in large-scale multiplayer games [Carlini and Lulli 2019; Miller and Crowcroft 2009].

Player behavior models can also benefit asset creation to improve player experience. [Durst et al. 2024] successfully created data-driven team movement models that allow AI opponents to simulate realistic human player movements. Level design and map

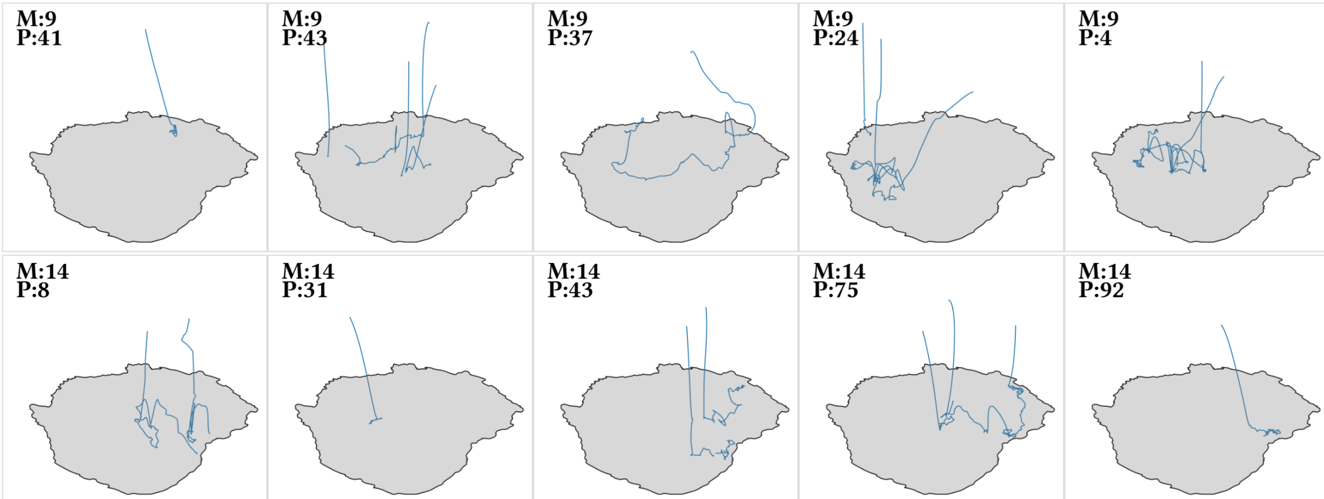

Figure 4: Player motion data visualization from the path breadcrumb telemetry data from Match 9 and 14.

making can be enhanced by using clustering techniques such as those explored by [Bauckhage et al. 2014] and design heuristics developed [Güttler and Johansson 2003] by examining player behavior [Mahlmann et al. 2010]. By using player data to drive design decisions and game mechanisms, developers can improve player experience and therefore increase chances of player retention and recruitment through current player recommendations.

Video game telemetry also allows for the unobtrusive collection of large swaths of human behavior data that can be used to help answer questions about teamwork and expertise. [Huang et al. 2017b] and [Huang et al. 2013a] used Halo data over a period of 7 months to analyze how players improve at the game and identify which players are more likely to be skilled at the game in the future. Information about behaviors such as breaks between games, other games played, and other macro level details about players' gaming habits allowed for insights into behaviors most linked to becoming experts. Work has also been done looking at how expertise interacts with team behaviors. [Drachen et al. 2014] examined movement behaviors of teams across different skill levels to find that expert teams have less variance in team positioning. By modeling player and team behavior, important insights can be made about the movement, coordination, and synchrony of teams to help teams maximize their potential while also providing clues to how team movement dynamics may translate into teams outside of esports [Ahmad et al. 2019; Habaraduwa et al. 2025; Mortelier et al. 2025].

3 Methodology

This section describes the dataset used in our analysis, the preprocessing steps applied to extract meaningful trajectories, and the measures of movement-based metrics computed for each player. We focus on spatial and temporal features derived from breadcrumb telemetry data collected during matches. These measures quantify patterns of player behavior and dynamics.

3.1 CoD Caldera Data

The data used in this study was released publicly by Activision in the summer of 2024 as part of the *Caldera data set* [Activision 2024], available at https://github.com/Activision/caldera. The dataset is distributed as a large hierarchical scene format using Universal Scene Description (USD), a scalable framework for representing complex 3D environments and assets [Pixar Animation Studios 2016]. The USD structure encodes both high-resolution map geometry and fine-grained player telemetry data across multiple nested levels of detail. This representation enables efficient parsing, selective loading, and semantic referencing of both spatial and temporal game elements.

The geometry component of the dataset contains approximately $\approx$ 17.5 million primitives, totaling over 2 billion points, and captures the full spatial layout of the Caldera map in a proxy and high-fidelity level of detail. The geometry data determined the map boundaries, points/zones of interest, and was used for rendering visualizations of player dynamics. The data also includes $24,365$ loot spawn points and transformation data for world locations. The player telemetry data includes breadcrumb locations and metadata for 4,187 players across 36 matches, with an average of $\mu = 116$ players per match. Each match lasted approximately 1,469.88 seconds (24.50 minutes) on average. A separate node contains over 1 million data points, marking the final player endpoints, which represent positions where players were eliminated or the match concluded. The endpoint data consists of positional data and time data that indicates the total time the represented player was in the game.

3.2 Data Processing

To prepare the raw telemetry data for analysis, we applied a series of preprocessing steps to ensure consistency, reliability, and spatial-temporal alignment across player trajectories. Each player's movement data consists of a time-ordered sequence of breadcrumb points, which include spatial coordinates (X, Y, Z), timestamps ($\approx$ every 2 seconds), and additional life metadata. We filtered the dataset

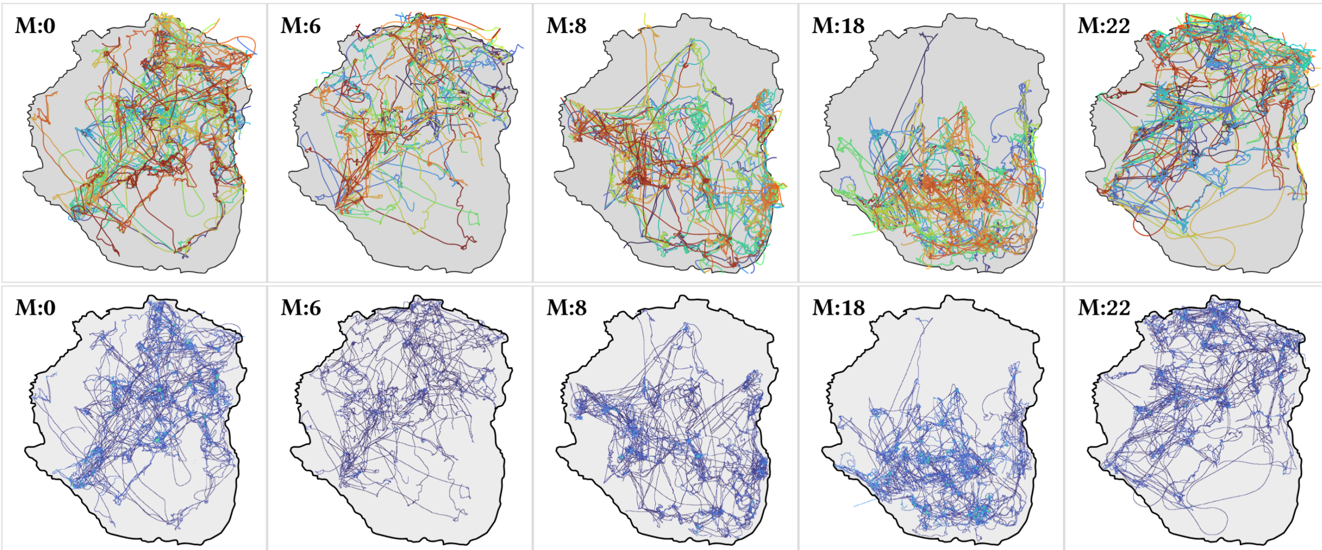

Figure 5: Player motion data from Match 0, Match 6, Match 8, Match 18, and Match 22. (Top) Path visualization from the path breadcrumb telemetry data, (Bottom) Heatmap from the same matches. (Additional figures in the supplementary document).

to include only complete player trajectories from valid matches. Match 26 (116 players) was removed since it lasted 6.31 minutes ($z = -12.3$), where the rest of the matches lasted $\mu = 25.02$ minutes. There was insufficient context to explain why match 26 was significantly shorter than the other matches, which did not justify its inclusion for analysis. 66 additional players were removed from the analysis due to either having no data, never deploying from the plane, or having less than 3 samples ($66 + 116 = 182$ total players removed), leaving 4121 players used for analysis.

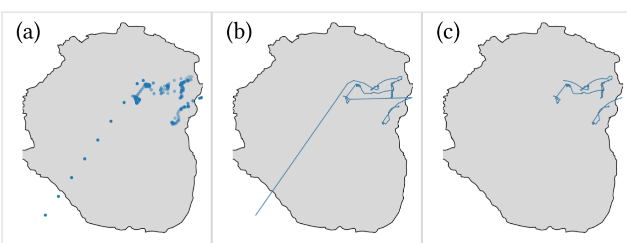

Figure 6: Player motion data from Match 0, Player 9 (final rank: 20th). (a) Original breadcrumb point data, (b) basic interpolation, (c) 2nd-degree spline with jumps and plane ride removed.

For each player's motion data, we filtered out trajectory segments with implausible movement artifacts, such as instantaneous long-distance displacements and/or unusually large time gaps between consecutive points indicative of gulag show downs, glitches, or teleportation. After removing these segments, we reconstructed player trajectories using spline interpolation to produce smooth and temporally consistent paths. For each segment, the original time-stamped X, Y, and Z coordinates were resampled using a cubic B-spline, generating uniformly spaced points along the trajectory. This approach preserved the overall movement pattern while reducing noise and avoiding sharp discontinuities caused by linear interpolation. Figure 6 illustrates this process, comparing raw breadcrumb points (a), linear interpolation with visible jumps (b), and the resulting smooth spline-based segment trajectories (c). Figure 4 shows some example player paths and matches (Figure 5). All paths for an entire match and paths/heatmaps for all matches can be found in the supplementary document.

3.3 Measures

To analyze player behavior in relation to in-game performance, we computed a set of movement and spatial metrics from the breadcrumbs. First, players were ranked within each match based on their survival duration, where rank 1 was assigned to the player with the longest total time in game. Since Warzone™ is a battle royale style game, this ranking metric serves as a representation of performance, independent of specific gameplay, actions, or kills. Players with lower valued ranks were considered high performers with 1 being the best possible rank. For each match, every player has a unique rank.

3.3.1 Movement Metrics. Player movement was calculated by measuring the *total distance* traveled over the course of a match. This was computed as the sum of 3D Euclidean distances between consecutive valid motion segments, excluding jump discontinuities and teleportation artifacts.

$$Distance = \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2} \quad (1)$$

Additionally, we segmented each match into temporal partitions to assess the evolution of movement behavior. These segments included the first and second halves (H1, H2), as well as quartiles (Q1–Q4) based on the total match duration. This allowed us to capture shifts in movement behavior as the game progressed, studying

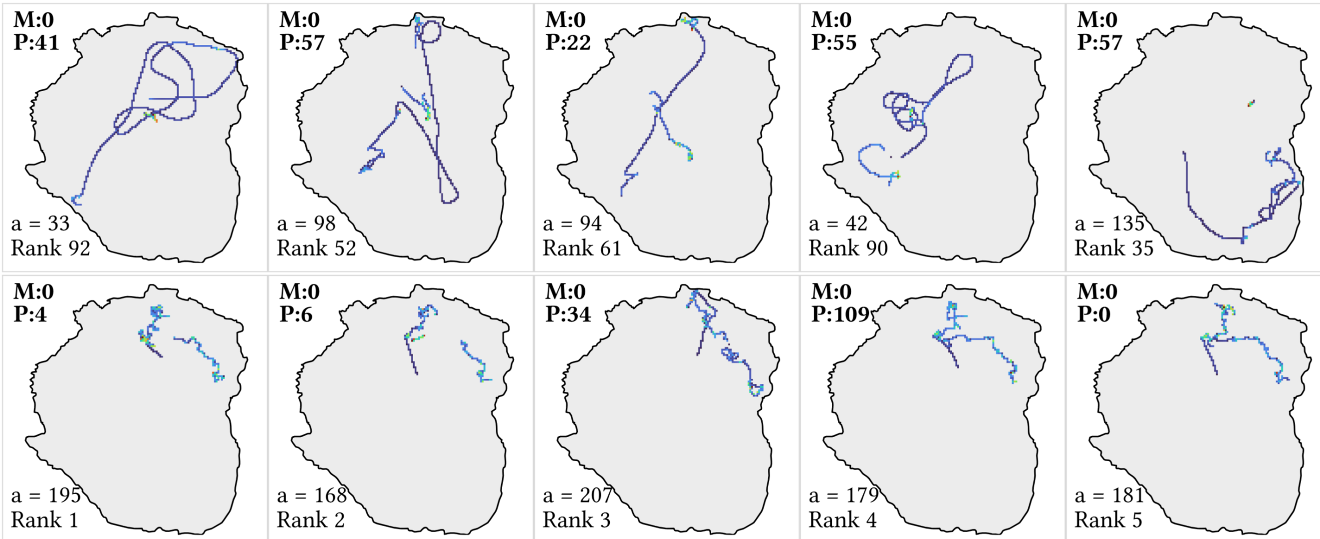

Figure 7: Heatmaps of player movement. (Top) Lower-ranked players with higher movement and fewer acceleration zero crossings *(a)*. (Bottom) Higher-ranked players with purposeful movement and dynamic behavior.

dynamic pacing and variability. We also performed a similar temporal segmentation using each player's end time to better align the analysis with their personal game duration and individual performance. To capture the dynamics of movement, we computed first, second, and third-order derivatives of position: *velocity*, *acceleration*, and *jerk*. Given a position vector $\boldsymbol{p}_i = (x_i, y_i)$, these quantities were derived as:

$$\boldsymbol{v}_i = \frac{p_{i+1} - p_i}{\Delta t}, \quad \boldsymbol{a}_i = \frac{v_{i+1} - v_i}{\Delta t}, \quad \boldsymbol{j}_i = \frac{a_{i+1} - a_i}{\Delta t} \tag{2}$$

We also computed the *zero-crossing rates* for velocity, acceleration, and jerk by counting the number of times the higher order derivative is equal to 0. These metrics quantify changes in direction, bursts of movement, or oscillatory behavior. These dynamics offer insight into the fluidity and responsiveness of a player's movement, such as sudden stops, starts, or erratic navigation patterns.

Total Path Tortuosity measures the total angular deviation in the player's path in the x-y plane across valid segments, quantifying how winding or circuitous the path is. Higher values indicate more turning, by summing the total angular change of the path. This quantity measures how much the path turns in total, where a higher value represents a more zig-zag high-turning path.:

$$\begin{aligned} T &= \sum_{i=0}^{N-3} \left| \theta_{i+1} - \theta_i \right|, \quad \textit{where} \\ \theta_i &= \arctan 2\left(y_{i+1} - y_i,\ x_{i+1} - x_i\right) \end{aligned} \tag{3}$$

3.3.2 Spatial Metrics. We generated 2D heatmaps (Figure 5 (bottom)) for each player using a fixed $N \times N$ grid over the map to analyze spatial-temporal movement. Each cell recorded the number of times the player's cubic B-spline position fell within its boundaries. We set $N = 150$ for these metrics. We derived the number of cells in the heatmap visited at least once by the player by summing the number of nonzero cells. The Shannon entropy of the heatmap distribution measures the unpredictability or uniformity of the player's spatial movements. Higher entropy indicates a more diffuse/less repeated movement pattern. To calculate this, we normalize the heatmap H to form a probability distribution: $p_{i,j} = \frac{H_{i,j}}{\sum_{i,j} H_{i,j}}$ if $\sum_{i,j} H_{i,j} > 0$, then compute Shannon entropy:

$$\text{Spatial Entropy} = -\sum_{i=1}^{N}\sum_{j=1}^{N} p_{i,j} \cdot \log_2(p_{i,j}) \quad \text{for} \quad p_{i,j} > 0 \tag{4}$$

We compute the variance of cell visits to quantify how unevenly a player distributes their movement across the map. Higher variance values indicate that movement is concentrated in a smaller number of regions, reflecting less spatial diversity.

3.3.3 Performance Metrics. We post-hoc separated players into high performing and low performing groups using a quartile split of each match and identified the first quartile as the high performers and the third quartile as the low performers. High performers were coded as 1, and low performers were coded as 3 to match the inverted scoring used for rank where lower values indicate higher performance. Using a quartile split to create binary groupings accounted for the unique nature of battle royale style games, where players do not all play the entire match. Since players can be eliminated as soon as the game starts, some of the lowest performing players were only in the game for seconds before being eliminated. Comparing their game data with the data from a player who played for over 20 minutes does not create a fair or meaningful comparison between the groups. The second quartile was also excluded to maintain consistency with the exclusion of the fourth quartile and to keep experimental groups at a similar size. Although our methodology is uncommon, discarding the second and fourth quartiles of players ensured that the data was meaningful and proportional. The logic behind the decision to employ this method would also apply to similar analyses of Fortnite, PlayerUnknown's Battlegrounds, and Apex Legends, which have the same game mechanics.

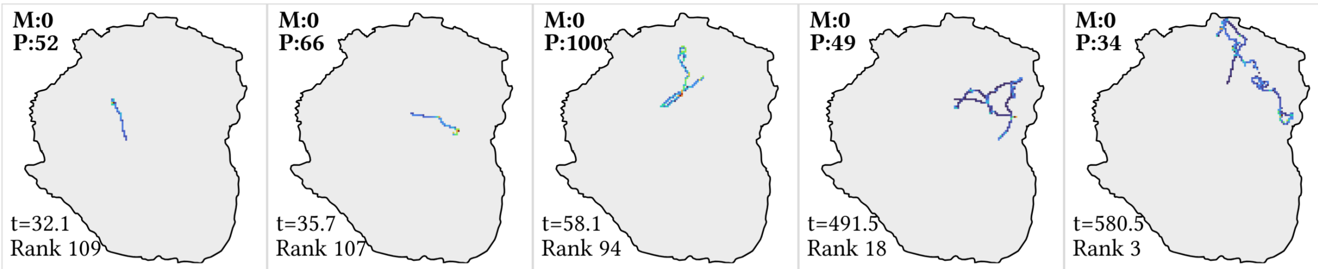

Figure 8: Visualization of player path tortuosity. Players with lower ranks tend to have straighter, more linear paths (low tortuosity), while higher-ranked players exhibit more complex and adaptive movement patterns (higher tortuosity).

4 Results

We analyze player behavior across three key dimensions of spatial endpoints and their proximity to loot, movement metrics distinguishing high- and low-performing players, and total distance traveled during each quarter of gameplay. These analyses provide insight into how positioning, mobility, and pacing relate to performance in Warzone™ matches.

Figure 9: Map of endpoint locations colored by time.

4.1 Endpoints and Loot Proximity

Figure 9 shows a map of player endpoints across the Caldera game environment with points color-coded by time to reveal temporal patterns in player deaths. This shows how endpoints cluster around major structures in the game. Figure 10 highlights spatial clustering by contrasting the top 2,500 and bottom 2,500 scoring endpoint times, showing where high performers and low performers ended their gameplay. This initial breakdown exposes emergent patterns in player behavior and gameplay outcomes, indicating that high- and low-performing deaths tend to cluster in different regions of the map. These visualizations suggest that spatial location is a key factor associated with player performance. Additional figures in the supplementary document further show endpoint distributions at varying thresholds (500, 1000, and 2500 deaths), supporting the observation of spatial clustering across performance tiers.

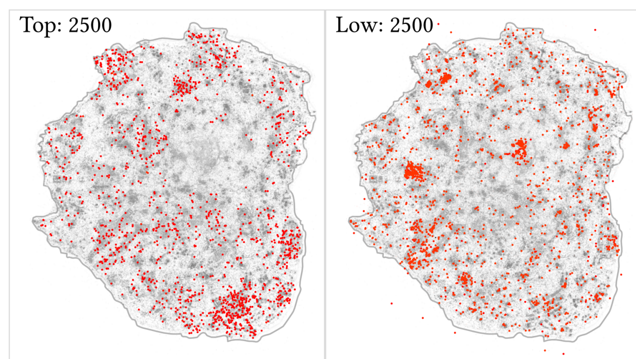

Figure 10: Map of top 2500 performers and low 2500 performers showing different clustering locations.

We analyzed the relationship between the total time in game from the endpoint locations and the minimum Euclidean distance to the nearest loot point. There was a statistically significant but very weak positive linear relationship ($r = 0.1$, $p = 0.033$, $z = 3.09$). This suggests that being near a loot point earlier in the match slightly increases the likelihood of being eliminated as players closer to the loot were in the game for less time. The weak relationship implies that proximity to loot does not strongly influence end location based on performance (rank/time in game), suggesting that other factors, such as movement patterns, risk avoidance, or broader gameplay movement, behavior, and dynamics, play a more significant role.

4.2 Movement Metrics of High and Low Performers

Figure 7 visualizes movement patterns across different player performance tiers. Lower-ranked/low performing players (top) show greater distance paths with fewer acceleration zero crossings, suggesting less adaptive movement. Higher-ranked/high performing players (bottom) exhibit purposeful exploration and frequent directional changes, indicating more responsive and tactical play. Figure 8 shows higher path tortuosity among top-ranked players, suggesting greater adaptability and responsiveness to in-game conditions, such as enemy encounters, terrain, and zone changes. Conversely, lower-ranked players tended to display more direct and predictable routes.

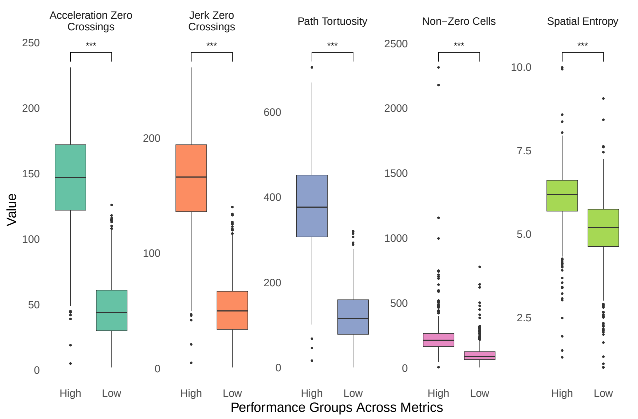

Figure 11: A comparative analysis of five different metrics. (Asterisks indicate statistically significant differences.)

Figure 11 and Table 1 analyze movement metrics to distinguish between high and low performing players using the performance groups described in section 3.3.3. The analysis focused on five metrics: *Acceleration Zero Crossings*, *Jerk Zero Crossings*, *Path Tortuosity*, *Non-Zero Cells*, and *Spatial Entropy*. The box plots in Figure 11 show that high-performance players consistently exhibited significantly higher values for Acceleration Zero Crossings, Jerk Zero Crossings, Path Tortuosity, and Non-Zero Cells, suggesting rougher, less direct, and inefficient movement patterns. These findings are consistent with the paths shown in Figure 7. This is supported by the correlation coefficients presented in Table 1, which show strong negative correlations between performance group and Acceleration Zero Crossings (-0.851), Jerk Zero Crossings (-0.850), Path Tortuosity (-0.832), and Non-Zero Cells (-0.521), while Spatial Entropy shows a moderate negative correlation (-0.492). The observed differences were consistently highly significant, with p-values frequently approaching zero, showing distinction between the performance groups.

4.3 Distances Each Quarter

Figure 12 and Table 2 present how player movement evolves over the course of a match, segmented into four equal-duration quarters based on the total time each player was in the match. The average distance traveled tends to decrease slightly in later quarters of gameplay, with lower-ranked players consistently covering more ground than their higher-ranked players. Correlation values remain positive across all quarters, with the strongest effect observed in the third quarter (Q3: 0.1746). While these correlations are relatively weak, the associated p-values approach 0.

Table 1: Performance groups are comprised of high and low ranked players with the high group (1) containing the top quarter of players from each match and low group (3) containing the 3rd quarter of players from each match.

Metric	Correlation with Performance Group
Acceleration Zero Crossings	-0.851
Jerk Zero Crossings	-0.850
Path Tortuosity	-0.832
Non-Zero Cells	-0.521
Spatial Entropy	-0.492

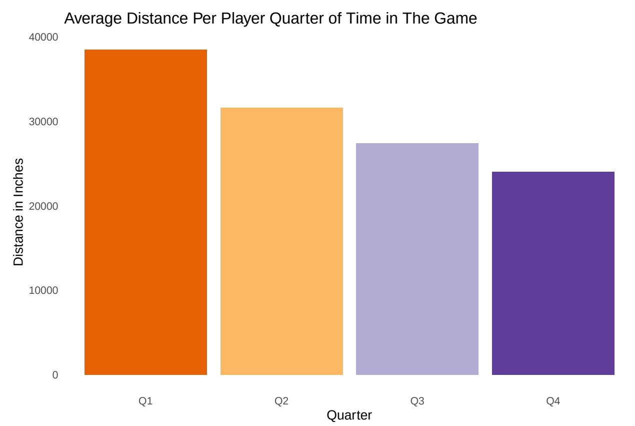

Figure 12: Visualization of the average distance covered by players during each quarter of a game. This provides insights into how player activity levels change throughout the different phases of a match.

Table 2: Quarters are determined based on each individual player's time in the game and the distance is normalized based on the time of each quarter. Players who had a worse rank moved more every quarter.

Total Distance Per Player Quarter	Correlation with Rank
Quarter 1 (Q1)	0.0797
Quarter 2 (Q2)	0.0944
Quarter 3 (Q3)	0.1746
Quarter 4 (Q4)	0.1534

Figure 13 visualizes individual player movement trajectories across each quarter of a match, with the top-ranked player highlighted in bold. This comparison reveals how high-performing players tend to show more compact and irregular motion patterns and how the pattern of motion changes during the game.

5 Discussion

The spatial and temporal analyses of player endpoints in the Caldera game environment reveal distinct patterns in player behavior and performance outcomes. The visualizations in Figures 9 and 10

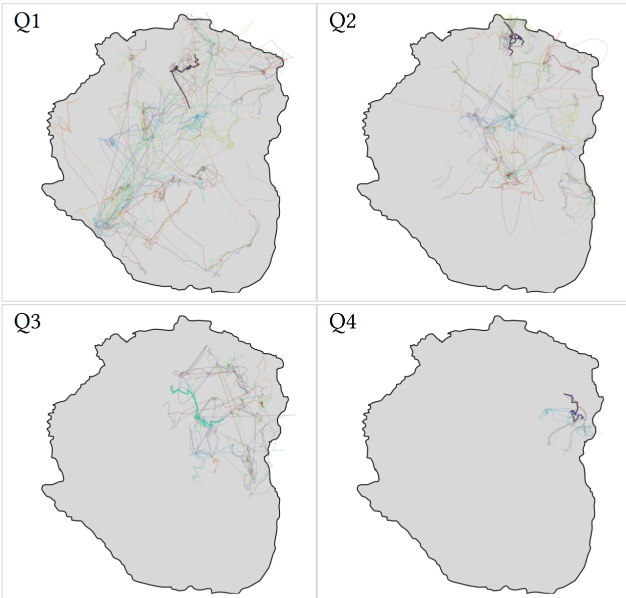

Figure 13: Motion of player for each quarter of a match, with the top ranked player in bold.

demonstrate that player deaths cluster around major structures, with high- and low-performing players exhibiting different spatial distributions. This spatial clustering suggests that map positioning plays a critical role in determining gameplay success, potentially influenced by factors such as resource availability, terrain features, or player density. The weak positive correlation between endpoint time and distance to loot points further indicates that proximity to loot is not a dominant factor in determining performance. Instead, broader gameplay dynamics, such as movement strategies and risk management, appear to have a more substantial impact on player outcomes.

The movement metrics analyzed in Figure 7, Figure 11, and Table 1 show behavior and dynamic differences between high and low performers. High performers exhibit more adaptive and tactical movement, characterized by frequent directional changes and purposeful exploration, as evidenced by higher zero crossings of acceleration and jerk. This is likely explained by higher performing players strategically seeking shelter, dodging enemies, searching for loot, and staying aware of their surroundings. Lower-ranked players exhibit less responsive movement patterns, characterized by greater path distances and fewer directional shifts, suggesting a lack of adaptability. Although smoother movements are typically associated with expertise and would therefore be expected of high performers, the findings here illustrate that the opposite is true for battle royale style games [Balasubramanian et al. 2015]. These findings show the interplay between spatial decision-making and movement dynamics in shaping player performance during gameplay in this genre.

Table 2 and Figure 13 show a weak but positive correlation between total distance traveled per player quarter and player rank, with the highest correlation observed in the third quarter (Q3: 0.1746). Across all quarters of gameplay, lower-ranking players tended to move more than higher-ranking players, further suggesting that excessive or inefficient movement may be linked to poorer performance. This pattern may reflect riskier play styles, increased exposure, or a lack of positional strategy, particularly in the mid-to-late stages of the match, where survival and map awareness become increasingly critical. This is only a possible explanation for the behavioral patterns and further context would be needed to confirm this interpretation.

The presented analyses and metrics are only a subset of the significant relationships present in this dataset. We also computed a wide range of position-based derivatives and spatial spreads, elevation, and variance across the whole match, halves, and quarters for both the match and the player's time in the game. Many of these showed interesting variations but were omitted from the discussion. The selected analyses provided valuable insights into player behavior that we considered most informative. For example, the total distance traveled was significantly negatively correlated with rank, but due to Warzone™ being a battle royale style game, higher ranked players were in the game longer and had more data points. Although this was a significant finding, it does not reveal anything about the players' behaviors, but rather demonstrates a characteristic of the genre. Additionally, many of the analyses were repetitive, especially for temporally partitioned metrics such as halves and quartiles, which would have the same correlational direction and similar strength. Therefore, their inclusion did not provide any additional significant meaning.

6 Limitations and Future Work

The analysis conducted in this study was constrained by several limitations inherent to the dataset. First, the dataset lacked contextual details about whether the players and the matches were solo or team-based. Because it was unclear whether the matches involved players on teams, since no information was available to identify team membership, we did not conduct team-based analyses. Instead, we assumed that all matches were played solo. This assumption may oversimplify the dynamics of gameplay, as evidenced by the dataset, which includes multiple time and spatial jumps, suggesting that some matches may have been team-based or influenced by special event matches. Future work can examine player behaviors to determine whether players coordinated or collaborated.

Information about player skill level and experience is also not included in the dataset. Due to this constraint, analyses were performed under the assumption that a player's rank in the game indicated their skill level. Every player who won their match was treated as equally skilled. However, CoD Warzone™ utilizes skill-based matchmaking, so some matches may have consisted of more skilled players than other matches. Therefore, our measure of skill may not provide an accurate assessment of a player's actual skill level in relation to players from different matches.

The dataset also omitted fine-grained details about player interactions and in-game actions, such as kills, damage taken, or loot collected. This data could enhance player performance metrics beyond just time in game and how these factors impact play style. Without the context of players' in-game actions, analyses were limited to broad explorations of movement.

Finally, the dataset's sampling frequency posed a significant constraint. Data points were recorded only every two seconds, which, given the fast-paced nature of the game, resulted in substantial data loss, particularly for micro-level movements. To mitigate this, we employed cubic B-spline interpolation to estimate player trajectories between recorded data points, partially addressing the gaps in temporal resolution. However, this approach may introduce artifacts, such as unnatural smoothing or distortions in movement patterns, particularly in scenarios involving abrupt changes in player behavior. This low temporal resolution hindered the ability to capture micro player behavior and dynamics during matches.

7 Conclusion

This work presents a data-driven framework for analyzing player behavior in large-scale competitive game environments using telemetry from Call of Duty®: Warzone's™ Caldera dataset. Reconstructing continuous trajectories from sparse breadcrumbs and computing interpretable movement and spatial metrics capture meaningful differences in player styles, pacing, and adaptability of various performance levels. Our findings demonstrate how spatial-temporal modeling of in-game behavior can reveal patterns tied to success and provide a foundation for deeper insights in game design, player modeling, and human behavioral analysis within virtual environments. As game data science continues to evolve, methods like these offer powerful tools for understanding the dynamics of competitive play at scale.

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